**Stochastic Processes Semester 1**

**Tutorial 1 – Solutions**

1. Recall that the moment generating function $\varphi(t)$ of a random variable $X$ is defined by

$$\varphi(t) = \mathbf{E}\left(e^{tX}\right)$$

and that for a nicely behaved random variable we have

$$\frac{d}{dt}\varphi(t)\bigg|_{t=0} = \mathbf{E}(X) \quad \text{and} \quad \frac{d^2}{dt^2}\varphi(t)\bigg|_{t=0} = \mathbf{E}\left(X^2\right).$$

Let us therefore first calculate $\varphi(t)$:

$$\begin{aligned}\varphi(t) &= \mathbf{E}\left(e^{tX}\right) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda+tk} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda\left(e^t-1\right)}.\end{aligned}$$

Now to find the moments, we differentiate the function $\varphi(t)$ using the chain rule:

$$\frac{d}{dt}\varphi(t) = \frac{d}{dt}(e^{\lambda\left(e^t-1\right)}) = \lambda e^t e^{\lambda\left(e^t-1\right)},$$

so $\mathbf{E}(X) = \frac{d}{dt}\varphi(t)|_{t=0} = \lambda$. Similarly,

$$\frac{d^2}{dt^2}\varphi(t) = \lambda e^{t+\lambda\left(e^t-1\right)}(1+\lambda e^t),$$

and so $\mathbf{E}(X^2) = \frac{d^2}{dt^2}\varphi(t)|_{t=0} = \lambda + \lambda^2$.

2.

$$\begin{aligned}\mathbf{E}\left(X^2\right) &= \sum_{k=0}^{\infty} k^2 \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} k^2 \cdot \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda e^{-\lambda} \sum_{k=1}^{\infty} k \cdot \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \left(\frac{\lambda^k}{(k-1)!}\right)' \\ &= \lambda e^{-\lambda} \left(\sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}\right)' = \lambda e^{-\lambda} \left(\lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}\right)' \\ &= \lambda e^{-\lambda} \left(\lambda e^{\lambda}\right)' = \lambda e^{-\lambda} \left(\lambda e^{\lambda} + e^{\lambda}\right) = \lambda^2 + \lambda.\end{aligned}$$

3. First, we calculate $\mathbf{E}(X)$ using integration by parts:

$$\mathbf{E}(X) = \int_0^{\infty} x\alpha e^{-\alpha x} dx = -xe^{-\alpha x}\Big|_0^{\infty} + \int_0^{\infty} e^{-\alpha x} dx = 0 - \frac{1}{\alpha} e^{-\alpha x}\Big|_0^{\infty} = \frac{1}{\alpha}.$$

Thus the formula holds for $n = 1$. Now we will show by induction that $\mathbf{E}(X^n) = n!/\alpha^n$. Assume this formula holds for the $(n-1)$st moment and now prove it for the $n$th. Again, use integration by parts and the induction hypothesis:

$$\mathbf{E}(X^n) = \int_0^{\infty} x^n \alpha e^{-\alpha x} dx = -x^n e^{-\alpha x}\Big|_0^{\infty} + \int_0^{\infty} n x^{n-1} e^{-\alpha x} dx = 0 + \frac{n}{\alpha}\mathbf{E}\left(X^{n-1}\right) = \frac{n \cdot (n-1)!}{\alpha \cdot \alpha^{n-1}} = \frac{n!}{\alpha^n}.$$

4. (i) Use the change of variables $y = (x-\mu)/\sigma$. Then $dx = \sigma dy$ and the integration limits for $y$ are also $-\infty$ and $\infty$. So

$$\begin{aligned}\mathbf{E}(X) &= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \int_{-\infty}^{\infty} (\sigma y + \mu) \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-y^2/2} \sigma dy \\ &= \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy + \sigma \int_{-\infty}^{\infty} y \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy = \mu + 0 = \mu\end{aligned}$$

since the first integral in the sum is the integral of the standard normal density (and has to be equal to 1 by the fact that it is a density) and the function in the second integral is odd. For the second moment, introduce the random variable $Y = (X-\mu)/\sigma$. Then $Y$ has the standard normal distribution and then

$$\mathbf{E}(X^2) = \mathbf{E}(\sigma Y + \mu)^2 = \sigma^2 \mathbf{E}(Y^2) + 2\sigma\mu \mathbf{E}(Y) + \mu^2$$

where $\mathbf{E}(Y) = 0$ (since the integral is odd) and

$$\begin{aligned}\mathbf{E}(Y^2) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 \cdot e^{-y^2/2} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y \cdot \left(-e^{-y^2/2}\right)' dy \\ &= -\frac{y}{\sqrt{2\pi}} e^{-y^2/2} \Big|_{-\infty}^{\infty} + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy = 0 + 1 = 1.\end{aligned}$$

Therefore

$$\mathbf{E}(X^2) = \sigma^2 + \mu^2 \quad \text{and} \quad \mathbf{Var}\,(X) = \mathbf{E}(X^2) - (\mathbf{E}(X))^2 = \sigma^2.$$

(ii) Now assume that $X \sim N(0,1)$. For $n$ odd, the function $y^n e^{-y^2/2}$ is also odd and therefore

$$\mathbf{E}(X^n) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^n e^{-y^2/2} dy = 0.$$

For even values of $n$, we use the integration by parts and induction:

$$\begin{aligned}\mathbf{E}(X^n) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^n e^{-y^2/2} dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^{n-1} \left(-e^{-y^2/2}\right)' dy \\ &= -\frac{1}{\sqrt{2\pi}} y^{n-1} e^{-y^2/2} \Big|_{-\infty}^{\infty} + (n-1) \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^{n-2} e^{-y^2/2} dy = 0 + (n-1)\mathbf{E}(X^{n-2}) \\ &= \ldots = (n-1)(n-3)\mathbf{E}(X^{n-4}) = \ldots = (n-1)(n-3) \cdot \ldots \cdot 3 \cdot 1.\end{aligned}$$

5. Since the density is a constant $C$ inside the triangle, the total integral is equal to a product of $C$ and of the area of the triangle and also equals 1 (draw a picture to see why). Therefore $C = 2/3$.

Now we find $f_Y(y)$. The edge of the triangle that connects point $(1,0)$ and $(0,3)$ satisfies the equation $x + y/3 = 1$. Clearly, $f_Y(y) = 0$ if $y < 0$ or $y > 3$. For $0 \le y \le 3$,

$$f_Y(y) = \int_0^{(1-y/3)} \frac{2}{3} dx = \frac{2(1-y/3)}{3}.$$

Similarly, $f_X(x) = 2(1-x)$, for $0 \le x \le 1$, and is 0 otherwise.

6.
$$\mathbf{E}(Y) = \int_0^3 y \cdot \frac{2(1-y/3)}{3} dy = 1$$
and
$$\mathbf{E}(X) = \int_0^1 x \cdot 2(1-x) dx = 1/3.$$
Further,
$$\mathbf{E}(XY) = \int_0^1 x \left( \int_0^{3(1-x)} y \cdot \frac{2}{3} dy \right) dx = \int_0^1 3x(1-x)^2 dx = \int_0^1 3x^2(1-x) dx = 1/4.$$

7. We define conditional densities only for values $0 \le x \le 1$ and $0 \le y \le 3$ and let them equal to 0 otherwise (since for other values the full density is 0). So,
$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{2/3}{2(1-x)} = \frac{1}{3(1-x)}$$
if $x \ge 0$, $y \ge 0$ and $x + y/3 \le 1$, and $f_{Y|X}(y|x) = 0$ otherwise. Similarly,
$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{2/3}{2(1-y/3)/3} = \frac{1}{1-y/3}$$
if $x \ge 0$, $y \ge 0$ and $x + y/3 \le 1$, and $f_{X|Y}(x|y) = 0$ otherwise. Further,
$$\mathbf{E}(X|Y=y) = \int_0^{1-y/3} x \cdot \frac{1}{1-y/3} dx = \frac{1-y/3}{2};$$
$$\mathbf{E}\left(X^2|Y=y\right) = \int_0^{1-y/3} x^2 \cdot \frac{1}{1-y/3} dx = \frac{(1-y/3)^2}{3};$$
and
$$\mathbf{Var}\left(X|Y=y\right) = \frac{(1-y/3)^2}{12}.$$